

Nishchith Rao

San Francisco, CA | Open to Relocation

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SUMMARY

Software Engineer with 2+ years of experience building scalable microservices and distributed backend systems in cloud-native environments. Proven experience in delivering event-driven architectures, software design and full-stack applications in production systems serving 10000+ users.

EXPERIENCE

Full Stack Developer | *Saayam For All - San Jose, CA*

August 2025 – Present

- Developing 4 scalable core microservices using Java (Spring Boot), Python and Go, deployed on AWS Elastic Kubernetes Service (EKS), supporting a platform with 10,000+ active users.
- Designed an event-driven real-time notification system using Apache Kafka (MSK), achieving < 200ms event propagation latency and 99.9% delivery reliability across distributed services.
- Built secure RESTful APIs with TypeScript and Node.js, integrated with AWS API Gateway, Cognito, Lambda and RDS. Developed React front-end components contributing to a successful 1.0 MVP release and improved the speed of feature rollout cycles.
- Implemented user profile deduplication logic and data validation workflows, improving cross-service data consistency and reducing redundant account creation.

Associate Software Engineer | *Capgemini Engineering - Bangalore, India*

February 2022 – July 2023

- Developed backend modules and network convergence features for Cisco's NCS 4K routing platform, part of core infrastructure used by major global carriers, ISPs and other Tier-1 operators for high density WAN and TCP/IP networking.
- Engineered optical fiber configuration integrations using Core Java, Hibernate and XML, contributing to a measurable 4.5% improvement in router processing performance.
- Built backend services with structured logging, fault tolerance, automated monitoring and supported GUI system upgrades for distributed telecom systems reducing production grade errors. Developed reusable TypeScript components/SDK modules consumed by multiple internal services, improving integration speed by 28%.
- Collaborated with cross functional and client engineering teams within Agile sprint cycles to translate evolving requirements into production ready implementations and rapid Figma prototypes.

Graduate Teaching Assistant | *Northeastern University - Boston, MA*

January 2024 – April 2025

- Supported instruction for graduate-level courses in Natural Language Processing (NLP) and Database Design, assisting 60+ students.
- Delivered lectures covering SQL optimization, transformer-based NLP pipelines, ML workflows and data modeling.
- Evaluated projects and provided technical feedback improving student implementation quality and system design understanding.

EDUCATION

Master of Science (MS) in Computer Science | *Northeastern University, Boston, MA*

September 2023 - May 2025

- Relevant coursework:** Algorithms, Distributed Systems, Design Principles, Database Management, Machine Learning

Bachelor of Science (BS) in Computer Science | *Presidency University, Bangalore, India*

August 2018 - June 2022

- Relevant coursework:** Data Structures, Object Oriented Programming (OOP), Operating Systems, Computer Networks

PROJECTS

ResuMatrix | [GitHub Link](#)

- Built an end-to-end MLOps pipeline for automated resume classification using SBERT embeddings for semantic matching and XGBoost with cosine similarity for filtering, improving job-description to resume match precision by 25%.
- Using Python, orchestrated scalable ETL data pipelines with Apache Airflow for automated ingestion, embedding generation and scheduled model retraining to mitigate data shift and model decay.
- Fine-tuned LLM-based ranking strategies with Gemini and Borda ranker to enhance requirements matching and mitigate data drift through scheduled retraining and performance monitoring.
- Deployed containerized services using Docker + Jenkins CI/CD on Google Cloud Platform (GCP), reducing storage costs by 40% while processing 400+ resumes per run. Monitored via MLflow and Grafana for experiment tracks, model performance and production visibility.

Fitness Tracker Application | [GitHub Link](#)

- Built a highly-available full-stack web application using Java (Spring Boot), Node.js, FastAPI and React designing 14 REST APIs for calorie tracking, goal management and diet analytics.
- Implemented Redis caching and secure authentication using Firebase Auth, reducing repeated database queries by 45% and cutting average API response latency from ~250ms to < 140ms. Designed UI with Javascript, React, Tailwind CSS/HTML for web delivery.
- Achieved 90%+ unit testing coverage using JUnit and Mockito, ensuring reliability and maintainability. Awarded 2nd Place at Capgemini Hack Week 2022 for its backend services architecture design and scalability.

Mars Terrain Exploration and Navigation Framework | [GitHub Link](#)

- Engineered an autonomous rover navigation framework using PPO and TD3 algorithms in OpenAI Gym simulated environments.
- Experimented with custom reward functions to maximize terrain coverage and energy efficiency to achieve a 70% optimized traversal efficiency across 150,000 simulations.
- Integrated LIDAR-based perception using OpenCV for obstacle detection and enhanced spatial awareness. Visualized agent learning dynamics using Unity ML-Agents and Matplotlib for real-time performance feedback.

Weather-Driven Smart Home Energy Consumption and Optimization | [GitHub Link](#)

- Engineered time-series forecasting feature pipelines for weather acquisition and temporal decomposition to simulate appliance scheduling using regression ML models. Built a minimization strategy that reduced household energy costs by 15% through peak-load optimization.
- Evaluated models using cross-validation and achieved 12% lower MAE and 9% lower RMSE compared to baseline regression models.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, C#, Golang

Frameworks & APIs: Spring Boot, Node.js, FastAPI, React, REST APIs, Microservices, Hibernate, Maven, Websockets, Django

Cloud & DevOps: Amazon Web Services (AWS - EC2, S3, Lambda, API Gateway, RDS), Git, Docker, Kubernetes (K8s), Jenkins

Data Engineering & ML: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, MLflow, Apache Spark, GitHub Copilot

Database & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka (MSK), Cassandra